



Nature's Engineers: How Animal Bodies Solve Problems



Life Science Unit, Lesson 1



Grade: 1st Grade



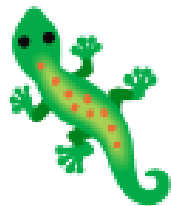
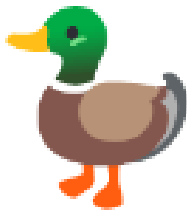
Duration: 45 minutes



Subject: Life Science — Animal Structures



Kit Lesson: No



Nature has been solving engineering problems for billions of years — can you figure out how?



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **identify** how specific animal body parts solve survival problems and **design** a simple invention inspired by one animal's structure.



The Big Idea

Engineers often get their best ideas from nature! When animals evolve a clever body part — hooked burrs for traveling, webbed feet for swimming, hollow bird bones for flying — human engineers have copied these ideas to solve real problems. This is called **biomimicry** (bio = life, mimicry = copying).



Did you know? In 1941, a Swiss engineer named George de Mestral went on a walk with his dog, Milka. When they got home, burrs were stuck all over the dog's fur — and on his own jacket! He looked at them under a microscope and saw tiny hooks. Eight years later, he invented **Velcro** — copying nature's design! Look at the Velcro on your shoes or backpack and you'll see those same tiny hooks.



Key vocabulary: structure, function, biomimicry, engineer, adapt, survive, invent

Materials

What You Need

- Paper and crayons (for the design challenge)
 - Index cards — *optional* (for making structure cards)
 - A magnifying glass — *optional but fun* (look at Velcro up close!)
- ✓ *Zero materials to buy — this lesson needs only paper and crayons!*

Part 1 — Animal Structures Investigation (15 min)

Read each animal card aloud together and figure out the problem it solves. For each one, ask: "What problem does this body part solve for the animal?"



Duck's Webbed Feet Problem: How do you swim fast? *Solution: Webbed feet act like paddles!*



Porcupine Quills Problem: How do you defend yourself without running? *Solution: Sharp quills deter predators!*



Gecko's Sticky Feet Problem: How do you climb smooth surfaces? *Solution: Special tiny hooks grip any surface!*



Cat's Retractable Claws Problem: How do you keep claws sharp AND walk quietly? *Solution: Retract them when not in use!*



Lotus Leaf Problem: How do you stay clean in muddy water? *Solution: Tiny bumps make water bead up and roll dirt away!*



Bonus: Burrs! Problem: How does a seed travel to a new place? *Solution: Tiny hooks catch on passing animals and fur!*

Discussion during Part 1:

For each card, ask:

- "What is the animal's problem?"
- "How does this body part help solve it?"
- "Can you think of any human tool that works the same way?"

Part 2 — Biomimicry Design Challenge (20 min)

Pick a Problem, Design a Solution!

Choose *one* of these everyday problems:

-  "My backpack keeps slipping off my shoulders."
-  "Water always gets my papers wet."
-  "I keep losing my socks."

Now think like an engineer: Which animal's body part could help solve this problem? Here are some ideas to get started — but encourage your student to come up with their own!

-  *Gecko feet* → non-slip backpack straps (tiny hooks on the inside)
-  *Lotus leaf* → waterproof paper coating (waxy bumps)
-  *Burrs* → reusable sock fastener (this is literally how Velcro was invented!)

Draw your invention. Label it. Name the animal that inspired it.

Step-by-Step Guide for Part 2:

1. Read the three problem choices aloud. Let your student pick one.
2. Look back at the animal structure cards. Ask: "Which animal's trick could help with this problem?"
3. On blank paper, draw the invention. Add labels: what's the problem? What animal inspired it? How does it work?
4. Share the invention with the family! Explain how it uses the animal's idea.

Discussion Questions

 **Question 1:** "Velcro was invented by a Swiss engineer who got burrs stuck on his dog's fur. Can you think of another everyday object that might have been inspired by nature?"

Suggested answer: So many! Shark skin inspired faster swimsuit fabric — tiny scales reduce drag in water. Kingfisher beaks inspired the nose of Japan's bullet train, making it quieter in tunnels. Lotus leaf inspired waterproof coatings on fabrics. Butterfly wings inspired screens that are readable in sunlight. Spider silk inspired super-strong materials. Nature has been solving engineering problems for billions of years!

 **Question 2:** "Your invention copies an animal's body part. Does that mean the animal is an engineer?"

Suggested answer: In a way — yes! Evolution is like a very slow engineering process. Animals that had body parts that worked better survived and had more babies, passing on those features. Over millions of years, this produced amazingly effective "designs." Scientists call this "natural selection" — nature's own engineering process. Animals didn't plan their designs, but the results are just as clever!



Extensions

- **Velcro Close-Up:** Find Velcro on a shoe or backpack. Look at it with a magnifying glass — can you see the tiny hooks? Compare the two sides: one has hooks, one has loops!
- **Shark Skin Swimsuits:** Look up how shark skin helped swimmers go faster in the Olympics. Tiny scales called "denticles" reduce drag — just like how a shark swims so smoothly!
- **Bullet Train Beak:** The nose of Japan's Shinkansen bullet train was redesigned to copy a kingfisher's beak. Look up pictures of both and compare!
- **Read Aloud:** *"Biomimicry: Inventions Inspired by Nature"* by Dora Lee, or *"What Do You Do With a Tail Like This?"* by Steve Jenkins
- **Nature Walk:** Go outside and find burrs, leaves with waxy coatings, or seeds with hooks. How many engineering ideas can you spot?



Parent/Teacher Notes

- **Zero materials to buy:** Paper and crayons only. The "structure cards" in Part 1 are just descriptions — read them aloud or write each one on an index card. Nothing to print or purchase.
- **The Velcro hook:** George de Mestral, 1941, burrs on his dog Milka → examined under microscope → 8 years of development → Velcro patent in 1955. If your student has Velcro on their shoes, pull it apart and look at each side under a magnifying glass — you can see the hooks and loops!
- **Gecko feet — the real science:** The actual mechanism is Van der Waals forces between millions of tiny hairs called *setae*. For Grade 1, "special tiny hooks" is a perfectly accurate simplification. Advanced students: search "gecko adhesive NASA" — researchers are studying gecko-inspired materials for space suits!
- **More biomimicry examples to share:** shark skin → Speedo swimsuits (some were banned from the Olympics for being too fast!), humpback whale fin bumps → more efficient wind turbine blades, spider silk → research into super-strong lightweight materials.
- **This connects to engineering design:** The design challenge in Part 2 directly fulfills NGSS K-2-ETS1 — students are identifying a problem, brainstorming solutions inspired by nature, and creating a design. That's real engineering design!
- **Lotus leaf detail:** The microscopic bumps on lotus leaves create a water-repellent surface — water beads up and rolls away, taking dirt with it. This is called the "lotus effect" and has inspired self-cleaning paint, fabrics, and glass.
- **For younger siblings:** Younger children can participate by matching emojis to the animal cards and drawing their own favorite animal's "superpower."



Student Worksheet — Nature's Engineers

Name: _____ Date: _____

Part 1 — Matching: Animal Structure to Problem

Draw a line from each animal to the problem its body part solves!



Duck's webbed feet



Porcupine quills



Gecko's sticky feet



Cat's retractable claws



Lotus leaf

Stays clean without washing

Climbs smooth surfaces easily

Swims fast like a paddle

Keeps claws sharp AND walks quietly

Defends without running away

Part 2 — Design Your Biomimicry Invention

Which animal inspired you? Draw your invention below. Label what the problem is, which animal inspired you, and how it works!

 Draw your invention here...

The problem I'm solving:

The animal that inspired me:

How my invention works:

Fill in the blank:

Biomimicry means _____ .

 My Invention!

My invention is called _____ .

It was inspired by _____ .



PARENT ANSWER KEY — Detach before giving worksheet to students

Part 1 — Matching Answers

Animal → Problem

-  Duck's webbed feet → **Swims fast like a paddle**
-  Porcupine quills → **Defends without running away**
-  Gecko's sticky feet → **Climbs smooth surfaces easily**
-  Cat's retractable claws → **Keeps claws sharp AND walks quietly**
-  Lotus leaf → **Stays clean without washing**

Biomimicry Inventions

-  Webbed feet → swim fins, paddles
-  Quills → hypodermic needles (inspired!), porcupine-inspired surgical staples
-  Gecko feet → non-slip grips, NASA space suit adhesives
-  Retractable claws → retractable pens, folding tools
-  Lotus leaf → waterproof fabric, self-cleaning glass
-  Burrs → Velcro (George de Mestral, 1955)

Fill in the Blank — Accepted Answers

- "**Biomimicry means ____.**" → *copying nature's ideas to solve human problems* (accept any equivalent phrasing: "copying animals," "learning from nature," "using animal ideas to make inventions")

Design Challenge — What to Look For

- Student identified a specific animal body part as their inspiration
- Drawing connects the animal structure to a human problem
- Student can explain in their own words how the invention copies the animal's design
- Invention is labeled with: problem, animal inspiration, and how it works

What Success Looks Like

- Student correctly matches at least four of five animal-to-problem pairs
- Student can explain: "Biomimicry means copying nature to solve a problem"
- Student's design clearly identifies an animal inspiration — even if the drawing is simple
- Student can tell you which animal inspired their invention and why



Bonus discussion answer: Other biomimicry examples your student might guess — shark skin → Speedo swimsuits, kingfisher beak → bullet train nose, humpback whale fins → wind turbine blades, spider silk → strong ropes and vests. Celebrate every connection they make to the natural world!

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Roots, Stems, Leaves, Flowers — What Each Part Does



Life Science Unit 1, Lesson 2



Grade: 1st Grade



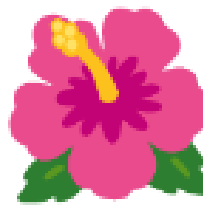
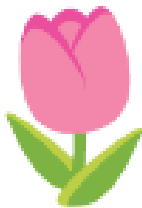
Duration: 45 minutes



Subject: Life Science — Plant Structures



Lesson: 2 of 6



Every part of a plant has a job — and plants figured out how to survive without a mouth, stomach, or legs!



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **identify** the four main plant parts and **explain** the survival function of each part.



The Big Idea

Plants are survival experts — but they do everything differently than animals. They can't chase food, run from danger, or drink from a stream. Instead, each plant part is perfectly designed for a specific job. Together, these parts help the plant get water, make food, and make more plants.



Key vocabulary: roots, stem, leaves, flower, photosynthesis, nutrients, absorb, pollinator, reproduce

Materials

What You Need

- A real houseplant, herb (basil, mint, parsley), or a weed from outside — *optional but great!*
- Paper and crayons (for the drawing activity)
- A cup of water (for the stem demonstration — see Teacher Notes)
- A celery stalk with leaves — *optional extension activity*
- Food coloring — *optional, for colored water celery demo*

✓ No plant? No problem — describe the parts from the drawings on this page!

Part 1 — Meet the Four Plant Parts (15 min)

Look at a real plant together, or study the descriptions below. For each part, ask: "What job does this part do to help the plant survive?"



Roots

Two jobs:

- (1) *Anchor* the plant in the soil so it doesn't blow away or fall over.
- (2) *Absorb water and minerals* from the soil — like drinking through a straw!

 *Root tip hairs are so tiny you need a magnifying glass to see them — but there can be millions!*



Stem

Two jobs:

- (1) *Support* the plant — holds leaves up toward the sun.
- (2) *Transport* water and nutrients from roots up to leaves, like a highway inside the plant.

 *Tiny tubes inside the stem called xylem carry water up. That's what makes celery change color when you put it in dyed water!*



Leaves

The plant's kitchen!

Leaves capture *sunlight* and use it — along with water from roots and carbon dioxide from air — to make *sugar* (food). This is called *photosynthesis*.

 *Tiny holes in leaves called stomata breathe in carbon dioxide and breathe out oxygen — the air we breathe!*



Flowers

The plant's way of making babies!

Flowers *attract pollinators* (bees, butterflies, hummingbirds) with bright colors and sweet nectar. Pollinators carry pollen from flower to flower so new seeds can form.

 *A flower's bright colors are basically an advertisement — "Free nectar here!" — for insects and birds.*

Discussion during Part 1:

For each part, ask:

- "What would happen to the plant if it had no roots? No leaves? No stem?"
- "Which part is doing the most important job? Can you decide?" (Trick question — they all depend on each other!)
- "What is the plant's 'kitchen'?" (Leaves!)



Part 2 — Plant Part Matching Activity (15 min)

On the worksheet, students draw lines matching each plant part to its job. Then they label a plant diagram.

Step-by-Step Guide for Part 2:

1. Go over the four plant parts together using the cards above.
2. Hand out the worksheet. Read the matching activity aloud together.
3. Students draw lines connecting each plant part to its function.
4. In the "Draw Your Plant" section, students draw a simple plant and label: roots, stem, leaves, flower.
5. Extension: Go outside and gently pull a weed from loose soil. Observe the root system! How long are the roots compared to the part above ground?



Discussion Question



Big Question: "A plant has no mouth or stomach. How does it get the energy to grow?"

Suggested answer: Plants make their own food using sunlight, water, and air — a process called photosynthesis. Leaves capture sunlight, roots absorb water from soil, and tiny pores in leaves take in carbon dioxide from the air. The plant combines these ingredients to make sugar, which is its food and energy. This is completely different from how animals eat — plants are more like solar-powered food factories than hungry hunters! This is why plants need sun and water, but animals need to eat other living things.



Extensions

-  **Colored Celery:** Put a celery stalk in a cup of water with several drops of food coloring. Wait 24 hours. Slice the stalk and look at the tiny colored dots — those are the *xylem* tubes that carry water up the stem!
-  **Weed Excavation:** Pull a weed gently from loose soil or a pot. How long is the root system? How does it compare to the plant's height?
-  **Leaf Test:** Cover part of a leaf with dark paper or foil for 3–5 days. The covered part will turn yellow because it can't get sunlight for photosynthesis. Uncovered parts stay green!
-  **Read Aloud:** "A Seed Is Sleepy" by Dianna Hutts Aston, or "From Seed to Plant" by Gail Gibbons
-  **Stomata Observation:** Press clear tape onto the underside of a large leaf, peel it off, and stick it to a white paper. Look at it through a magnifying glass — can you see the tiny pores (stomata)?



Parent/Teacher Notes

- **Photosynthesis for Grade 1:** You don't need to use the word "photosynthesis" unless your student enjoys big words. The core concept: "Leaves are the plant's kitchen — they use sunlight to make food." That's enough! If they're curious, add: the plant mixes sunlight + water + air to make sugar.
- **Colored celery tip:** Use blue or red food coloring for best visibility. This works beautifully in 1–2 days. Cut the stalk from the bottom at an angle for faster uptake. This is one of the most memorable plant biology demos — highly recommended!
- **Why leaves are green:** Plants are green because of chlorophyll, a molecule that absorbs red and blue light but reflects green light. The plant uses red and blue light for photosynthesis. This is fun to mention if your student asks, but not required for Grade 1.
- **Roots — more depth for curious kids:** Root hairs are single-cell extensions that dramatically increase surface area for water absorption. A single rye plant can have 14 billion root hairs! Taproots (like carrots) go deep; fibrous roots spread wide. Both strategies work — different environments favor different types.
- **Flowers are not just for beauty:** Remind students that flowers aren't there because they look nice to us — they evolved to attract specific animals. Their beauty is functional! (This connects to Lesson 4 on pollinators.)
- **If you have no plants:** Draw a simple plant on paper before the lesson, or use an image search. Even describing the parts from the cards on this page is enough for the activity.
- **Connection to Lesson 4:** This lesson introduces flowers and pollinators. Lesson 4 dives much deeper into how plant structures attract specific pollinators — a great "coming soon!"



Student Worksheet — Plant Parts and Their Jobs

Name: _____ Date: _____

Part 1 — Matching: Plant Part to Its Job

Draw a line from each plant part to what it does!



Roots



Stem



Leaves



Flower

Attracts bees and butterflies to make seeds

Captures sunlight to make food (the plant's kitchen!)

Holds the plant up and carries water upward

Soaks up water from soil and anchors the plant

Part 2 — Label the Plant

Draw a plant below with all four parts. Label each part: **roots**, **stem**, **leaves**, **flower**.



Draw your plant here — remember all four parts!

Part 3 — Fill in the Blanks

1. The _____ capture sunlight and make food for the plant.
2. The _____ drink up water from the soil.
3. The _____ carries water from roots up to the leaves.
4. The _____ attracts bees and butterflies to make new plants.

Part 4 — Think About It!

A plant has no mouth or stomach. How does it eat?



PARENT ANSWER KEY — Detach before giving worksheet to students

Part 1 — Matching Answers



Plant Part → Job



Roots → Soaks up water from soil and anchors the plant



Stem → Holds plant up and carries water upward



Leaves → Captures sunlight to make food



Flower → Attracts bees and butterflies to make seeds



Fill in the Blanks

1. **Leaves** capture sunlight
2. **Roots** drink up water
3. **Stem** carries water upward
4. **Flower** attracts pollinators

Part 4 — Think About It! Accepted Answers

- Plants make their own food using sunlight, water, and air (photosynthesis)
- Accept any phrasing that mentions: leaves, sunlight, water — the plant makes its own food
- "The leaves catch sunlight and use it to make sugar/food" is a full credit answer

Part 2 — Plant Drawing — What to Look For

- Plant drawing includes roots below the ground line (or below soil)
- Stem is visible connecting roots to the above-ground parts
- Leaves are attached to stem
- Flower (or at minimum a bud) is present at the top or on a branch
- All four parts are labeled correctly



What Success Looks Like

- Student correctly matches all four parts to their functions
- Student can explain: "Leaves make food using sunlight" (photosynthesis in student's own words)
- Student's plant drawing includes all four labeled parts
- Student understands that plant parts work together — roots get water, stem moves it, leaves use it



Science accuracy note: Plants don't truly "eat" like animals — they *make* their food through photosynthesis. If your student says "leaves eat sunlight," gently redirect to "leaves use sunlight to make food." The distinction matters!

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How Animals Protect Themselves

Life Science Unit 1, Lesson 3



Grade: 1st Grade



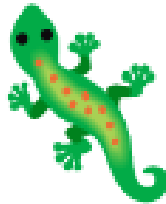
Duration: 45 minutes



Subject: Life Science — Animal Structures



Lesson: 3 of 6



Every animal has its own way of staying safe — and nature has invented some incredible solutions!



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **compare** at least four different structural defenses animals use and **design** a protection structure inspired by one.



The Big Idea

In nature, being eaten is a very real problem. Animals have evolved amazing body structures — called **structural defenses** — to help them avoid predators. These structures are built right into the animal's body and don't require any learning or planning. Let's explore five clever defense strategies!



Key vocabulary: defense, predator, prey, camouflage, armor, aposematism, spine, shell, structure

Materials

What You Need

- A toy figure (action figure, small animal toy, or even a LEGO minifigure)
- Craft supplies from around the house: tape, cardboard, foil, cotton balls, bubble wrap, rubber bands, fabric scraps — anything you have
- Paper and crayons (for the design worksheet)

✓ No toy figure? Just draw on paper — the design challenge works either way!

Part 1 — Five Defense Structures (15 min)

Study each defense strategy. For each one, ask: "What is the animal protecting itself from? How does this structure help?"



1. Hard Shell

Turtle, snail, hermit crab, clam

How it works: A rigid shell covers the animal's soft body. When danger comes, the animal retreats inside — it's like carrying a shield and a house all in one! The shell is made of bone or minerals, making it very hard to bite through.



2. Spines and Quills

Hedgehog, sea urchin, porcupine

How it works: Sharp spines stick out from the body. A predator that tries to bite gets poked — painfully! Porcupine quills even have tiny backwards hooks that make them harder to remove. The animal curls into a ball to protect its soft belly.



3. Camouflage Coloring

Stick insect, flounder, arctic fox, leaf-tailed gecko

How it works: The animal's colors and patterns match its surroundings perfectly. A predator looks right at the animal and doesn't see it! The flounder can even change its pattern to match different seafloors.



4. Warning Colors

Poison dart frog, monarch butterfly, yellow jacket

How it works: Bright, bold colors — red, orange, yellow, black — signal to predators: "I am dangerous, sick, or taste terrible! Don't eat me!" Predators that have learned this lesson leave these animals alone.

5. Armor Plating

Armadillo, pangolin, beetle

How it works: Overlapping plates of hard material (bone for armadillos, keratin — the same as your fingernails — for pangolins) cover the animal's back and sides. The armadillo can curl into a complete ball so no soft parts are exposed. Beetles have a hard outer "shell" made of their exoskeleton called an elytra.

Discussion during Part 1:

- "Which defense do you think is most effective? Why?"
- "Can you think of a predator that would be foiled by each type?"
- "If you were a small animal, which defense would you want?"

Part 2 — Design a Protection Gadget (20 min)

Engineering Challenge: Protect Your Figure!

Pick *one* defense strategy from above. Design a "protection gadget" for a toy figure (or draw one on paper) inspired by that animal's structure.

Your gadget must:

- Be inspired by a real animal's structural defense
- Physically protect the toy figure (or the drawn character) from a pretend threat
- Be labeled with: what it does, which animal inspired it, and how it works

Ideas to get started:

- 🐢 *Turtle shell*: Cut a dome shape from cardboard and attach it to the figure's back
- 🦔 *Porcupine quills*: Tape toothpicks or craft sticks around the figure so it's poky to grab
- 🐉 *Armor plates*: Wrap overlapping pieces of foil or cardboard (like scales) around the figure
- 🌿 *Camouflage*: Decorate the figure with markers or materials to match a specific background (grass, sand, bark)

Step-by-Step Guide for Part 2:

1. Read all five defense types together and discuss which is most interesting.
2. Student picks one defense to copy for their toy figure (or drawing).
3. Using available craft materials, build or draw the protection gadget.
4. Label the design: *Which animal inspired it? What does the protection do? How does it work?*
5. Optional: Test it! Gently poke or squeeze the figure and see if the protection works.



Discussion Question



Big Question: "Warning colors seem risky — they make you *more* visible. Why would that ever be a good survival strategy?"

Suggested answer: Warning colors (called aposematism) only work if the predator has already learned that those colors mean danger. Bright red and black on a monarch butterfly signals that it is poisonous from the milkweed it ate as a caterpillar. A bird that eats one monarch and gets sick will avoid all orange-and-black butterflies forever. So bright colors are actually a very smart strategy — they only require one bad experience to teach the whole predator population to stay away. It works so well that many harmless animals have evolved to copy the warning colors of dangerous ones — a trick called mimicry. The harmless milk snake has the same red-yellow-black bands as the venomous coral snake!



Extensions

-  **Camouflage Game:** Cut small pieces of green, yellow, and brown paper. Place them all on a piece of grass or a patterned fabric. From a few feet away, which ones are hardest to spot? That's camouflage in action!
-  **Mimicry Research:** Look up the "milk snake vs. coral snake" rhyme: "*Red on yellow, kill a fellow. Red on black, friend of Jack.*" Why would a harmless snake evolve to look like a deadly one?
-  **Flounder Video:** Search online for a video of a flounder changing color to match the seafloor. It's remarkable — they can match checkerboard patterns!
-  **Read Aloud:** "*What Do You Do When Something Wants to Eat You?*" by Steve Jenkins
-  **Pangolin Connection:** Pangolins are the world's most trafficked mammals — they're hunted for their scales. Their armor, which evolved to protect them from lions and leopards, is no match for human poachers. Discuss: how can a structural defense fail?



Parent/Teacher Notes

- **The warning color discussion:** "Aposematism" is a wonderful vocabulary word for curious Grade 1 students, but the concept is what matters. The key insight: warning colors are a population-level strategy, not individual. One individual has to get eaten to "teach" the predator. But the whole species benefits from that lesson spreading through predator behavior. This is sophisticated evolutionary thinking — but first graders can grasp it with the right framing: "One butterfly had to get eaten so all its orange-black butterfly friends would be safe."
- **Mimicry vs. camouflage:** These are easy to confuse. Camouflage = blend in, look like nothing. Warning colors = stand out, look dangerous. Mimicry = look like something else that IS dangerous (when you're actually harmless). All three are structural strategies shaped by evolution.
- **Armor plates science:** Pangolin scales are made of keratin — the same protein as your fingernails and hair. A pangolin rolled into a ball is so strong that lions have been observed batting them around for minutes and giving up. The scales are interlocking and can flex when the animal curls. Remarkable!
- **Craft material ideas:** Bubble wrap = soft armor. Toothpicks or craft sticks = quills. Crinkled foil = armor plates. Green yarn or paper shreds = camouflage. Be creative! The engineering challenge is open-ended by design.
- **Younger siblings:** Can participate in the "which defense would you pick?" discussion and help with the craft activity.



Student Worksheet — Animal Defense Structures

Name: _____ Date: _____

Part 1 — Five Defenses: Fill in the Animal!

Fill in the name of the defense each animal uses!

1. A turtle uses a _____ to protect itself.
2. A porcupine uses _____ to protect itself.
3. A stick insect uses _____ to hide from predators.
4. A poison dart frog uses _____ to warn predators to stay away.
5. An armadillo uses _____ to protect its soft body.

Part 2 — Design Your Protection Gadget

Pick one animal defense from above. Design a protection gadget for your toy figure inspired by that defense. Draw it below and label it!



Draw your protection gadget here...

The animal that inspired me:

The defense type (e.g. hard shell, quills):

How my gadget protects the figure:

Part 3 — Think About It!

Why do *warning colors* work as a defense even though they make the animal *more* visible?



PARENT ANSWER KEY — Detach before giving worksheet to students

Part 1 — Fill in the Defense



Answers

1. Turtle → **hard shell**
2. Porcupine → **quills / spines**
3. Stick insect → **camouflage**
4. Poison dart frog → **warning colors**
5. Armadillo → **armor plating**



Think About It! Answer

Warning colors work because predators *learn* to avoid them after one bad experience

The colors advertise danger or bad taste to any predator that has been warned before

Accept: "because the predator gets sick if it eats them and remembers not to do it again"

Part 2 — Design Challenge — What to Look For

- Student clearly identified one of the five defense strategies as inspiration
- Drawing shows a physical protection structure (not just decoration)
- Student can explain in words how the gadget mimics the animal's defense
- Labels include: animal inspiration, defense type, how it works



What Success Looks Like

- Student can name all five defense types and give one animal example for each
- Student's design challenge connects animal structure to protection function
- Student can explain warning colors: "predators learn to avoid them"
- Bonus: student can distinguish between camouflage and warning colors



Fun conversation starter: Ask: "Which defense would work best against a shark? Against a hawk? Against a cat?"
Different predators are defeated by different defenses. A camouflaged fish fools a diving bird but might not fool an echolocating dolphin!



How Plants Attract Pollinators

Life Science Unit 1, Lesson 4



Grade: 1st Grade



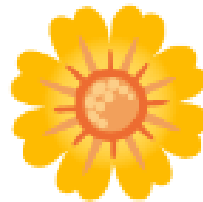
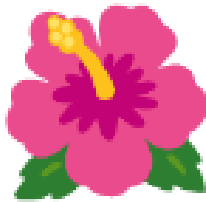
Duration: 45 minutes



Subject: Life Science — Plant Structures



Lesson: 4 of 6



Flowers are nature's billboards — every color, shape, and smell is an advertisement aimed at a specific visitor!



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **explain** how specific plant structures — color, smell, nectar, and shape — attract specific pollinators, and **design** their own "perfect flower" matched to a pollinator.



The Big Idea

Flowers don't bloom to be beautiful for us — every feature of a flower evolved to attract a specific pollinator. Color, shape, scent, and the location of nectar are all structural features designed (by evolution) to lure exactly the right animal. When that animal visits, it picks up pollen and carries it to another flower — and new seeds can form. It's a partnership: the plant gets pollinated, the visitor gets food.



Key vocabulary: pollinator, pollen, nectar, pollination, structure, attract, reproduce, seed

Materials

What You Need

- Paper and crayons or markers (for the flower design activity)
- Colored pencils — *optional, for more detailed flower design*

✓ This lesson is paper and crayons only — but the ideas are BIG!

Part 1 — Plant-Pollinator Matching (15 min)

Each flower type below has specific features that attract a specific pollinator. Study each pairing and discuss *why* the match works.

Red Tubular Flowers → Hummingbirds



Color: Bright red (hummingbirds can see red well — bees cannot!)

Shape: Long, narrow tube fits a hummingbird's long beak perfectly

Scent: Usually no smell — hummingbirds rely on sight, not smell

Nectar: Deep inside the tube, where only a long beak can reach



White Night-Blooming Flowers → Moths



Color: White or very pale — highly visible in moonlight

Shape: Often open and accessible

Scent: Very strong, sweet fragrance released at night — moths navigate by smell

Nectar: Available at night when moths are active



Blue and Purple Flat Flowers → Bees



Color: Blue or purple — bees see UV light and are especially attracted to these colors

Shape: Wide, flat, or platform-shaped — bees need a landing spot

Scent: Often sweet and pleasant

Nectar: Easy to access from the center of the flat flower



Large Pale Flowers with Rotting Smell → Flies and Beetles



Color: Pale, brownish, or muddy — mimics rotting flesh

Shape: Often large and deep — traps the insect briefly

Scent: Smells like rotting meat or dung — attracts insects that normally visit carrion

Nectar: Sometimes none — the flower tricks the insect!



Discussion during Part 1:

- "Why would a flower be red instead of blue? Who is it advertising to?"
- "What would happen if a bee visited a hummingbird flower? Would it work?"
- "Can you think of flowers you've seen outside? What pollinator do you think they attract?"

Part 2 — Design Your Perfect Flower (20 min)

Engineering Challenge: Design a Flower for a Pollinator!

Pick *one* pollinator:  Bee,  Hummingbird,  Butterfly,  Mosquito, or  Beetle.

Design your "perfect flower" that would attract *only* your chosen pollinator. Think carefully about:

-  **Color** — What color is your flower? Why would your pollinator like that color?
-  **Shape** — Is it a tube, a flat platform, a cup, a bell? Why does the shape suit your pollinator?
-  **Smell** — What does your flower smell like? Sweet? Fruity? Rotting? Nothing?
-  **Nectar location** — Where is the nectar hidden? Can your pollinator reach it?

Draw your flower and label all four features!

Step-by-Step Guide for Part 2:

1. Review the four pollinator-flower matchings together.
2. Student picks one pollinator and thinks about what would attract it most.
3. Draw the flower on the worksheet, making deliberate choices about color, shape, and smell.
4. Label: color, shape, smell, and nectar location.
5. Student explains their design: "I picked this color because... I made this shape because..."

Discussion Question

 **Big Question:** "A flower that smells like rotting meat seems disgusting. What kind of pollinator would visit that?"

Suggested answer: Flies and some beetles! The titan arum — nicknamed the "corpse flower" — is famous for this strategy. It heats itself up to near human body temperature and produces a powerful rotting flesh odor to attract carrion flies, which are insects that normally lay their eggs in dead animals. The flies enter the flower looking for a place to lay eggs, get dusted with pollen, and then leave disappointed (there's no actual meat) and carry the pollen to another corpse flower. Some orchids are even sneakier: they look and smell like female bees, tricking male bees into trying to mate with them — and picking up pollen in the process. Plants have evolved remarkable deception strategies to get pollinated without offering any reward at all!



Extensions

-  **UV Flower Patterns:** Bees can see ultraviolet light, which humans cannot. Many flowers have "nectar guides" — UV patterns that act like runway lights pointing toward the nectar. Search "flowers in UV light" to see images of flowers that look plain to us but have vivid bull's-eye patterns for bees!
-  **Garden Survey:** Walk outside and find 3–5 different flowers. For each one, guess: what pollinator is this flower "advertising to"? Check the color, shape, and smell. Then watch and see if you're right!
-  **Bat Pollinators:** Banana plants are pollinated by bats! Bat-pollinated flowers are usually pale (visible at night), produce very strong nectar, and bloom at night. Mangoes, figs, and durian also depend on bats.
-  **Read Aloud:** "*The Bee Book*" by Charlotte Milner, or "*Flower Talk: How Plants Use Color to Communicate*" by Sara Levine
-  **Bee Vision Art:** Draw a flower the way a human sees it, then draw the same flower the way a bee would see it (with UV patterns visible). Think about what new features you'd add for the bee version!



Parent/Teacher Notes

- **Pollination basics for Grade 1:** Pollen is a fine powder made by flowers. When it travels from one flower to another (carried by a pollinator), it allows a seed to form inside the flower. That seed can grow into a new plant. Pollinators get a meal of nectar (sugar water) or pollen (protein), and the plant gets its pollen delivered. It's a mutual benefit — usually!
- **The deceptive flowers discussion:** The corpse flower (titan arum) and bee orchids are fascinating but require care in presentation for Grade 1. The key idea: *some plants lie*. They offer a false promise (fake rotting meat, fake female bee) to trick an insect into carrying pollen. The insect gets nothing, but the plant's needs are met. This is a clever extension of the "advertisement" idea.
- **Bees and UV light:** Honeybees have three types of color receptors sensitive to UV, blue, and green — but not red! This is why red flowers typically don't attract bees (they look black or very dark to bees). Hummingbirds can see red, which is why hummingbird flowers are typically red or orange. This is a lovely example of how plant structures evolved to target specific visitors.
- **Connecting to Lesson 2:** Flowers were introduced in Lesson 2 as the "reproductive" part of the plant. This lesson deepens that — how exactly do flowers ensure their pollen travels? The answer is elaborate structural engineering aimed at specific visitors.
- **Night-blooming flowers note:** Moonflowers, evening primrose, night-blooming jasmine, and the queen of the night cactus are all moth-pollinated. Their scent is released at night when moths fly. Many are white because white reflects moonlight — another structure perfectly matched to function.



Student Worksheet — Flowers and Their Pollinators

Name: _____ Date: _____

Part 1 — Matching: Flower to Pollinator

Draw a line from each flower type to the pollinator it attracts!



Red, long tubular flower



White flower with strong night scent



Blue/purple flat flower



Large pale flower, smells like rotting meat



Flies and beetles



Bees



Hummingbirds



Moths

Part 2 — Design Your Perfect Flower

Pick a pollinator and design a flower just for it! Label all four features.

My pollinator is: _____



Draw your perfect flower here...



Color — and why:



Shape — and why:



Smell — and why:



Nectar location:



Part 3 — Think About It!

Why do flowers come in so many different shapes and colors?



PARENT ANSWER KEY — Detach before giving worksheet to students

Part 1 — Matching Answers



Flower → Pollinator



Red tubular → **Hummingbirds** 



White night-blooming → **Moths** 



Blue/purple flat → **Bees** 



Pale, rotting smell → **Flies and beetles** 



Part 3 — Think About It!

Different shapes and colors attract different pollinators

Each flower "advertises" to the specific animal that can best carry its pollen

Accept: "because different animals like different colors/smells"

Full credit answer: mentions that structure matches the pollinator

Part 2 — Design Challenge — What to Look For

- Student identified a specific pollinator before designing
- All four features are labeled: color, shape, smell, nectar location
- Features are *consistent* with what would attract the chosen pollinator (e.g., if they chose "bee," blue/purple colors and flat shape make sense)
- Student can explain the reasoning behind each choice



What Success Looks Like

- Student correctly matches all four flower-pollinator pairs
- Student's flower design shows intentional structural choices connected to the chosen pollinator
- Student understands that flower features are "advertisements" targeted at specific visitors
- Bonus: Student can explain why a hummingbird flower and a bee flower look different



Extended discussion: Ask "If all the bees disappeared, what would happen to blue and purple flowers?" (They'd have trouble reproducing — fewer seeds. Over many generations, they might evolve to attract a different pollinator or go extinct.) This is why bee populations matter so much — they're partners in plant reproduction for hundreds of thousands of plant species!

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Animal Structures for Getting Food



Life Science Unit 1, Lesson 5



Grade: 1st Grade



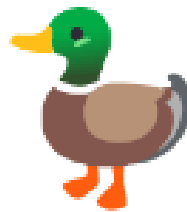
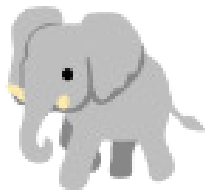
Duration: 45 minutes



Subject: Life Science — Animal Structures



Lesson: 5 of 6



Every animal has a specialized tool kit for getting food — no two are exactly alike!



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **match** animal body structures to their food-getting function and **explain** why each structure suits that specific diet.



The Big Idea

An animal's body parts are perfectly matched to what it eats. A meat-eater has different tools than a plant-eater. A filter-feeder has different tools than a seed-cracker. Over millions of years, animals with the best-matched body parts for their food source survived and passed on those features. The result: an amazing variety of specialized eating tools — beaks, tongues, trunks, claws, bills, and more!



Key vocabulary: beak, talon, trunk, filter, prehensile, specialized, diet, structure, predator, prey

Materials

What You Need

- Paper and crayons (for the drawing activity)
- Optional household items for a "beak challenge" extension: tweezers (eagle beak), a spatula (duck bill), a toothpick or straw (anteater tongue), a spoon (scooping/filter feeding) — and things to pick up: small marshmallows, rice, water with floating bits

✓ *Core lesson needs only paper and crayons. The kitchen challenge is a fun bonus!*

Part 1 — Six Amazing Food-Getting Structures (15 min)

Study each animal and its specialized body part. Ask: "How does this body part help the animal get its specific food?"



Eagle

Talons + hooked beak

Talons (curved, sharp claws) grip and hold prey tightly — even a struggling fish can't escape. The **hooked beak** tears meat into pieces small enough to swallow. Eagles can see 4–8 times more sharply than humans, which helps them spot prey from far away.

 **Eats:** fish, rabbits, squirrels, snakes



Elephant

Trunk (elongated nose + upper lip)

The **trunk** can reach high branches no other land animal can access, strip bark from trees, suck up water (holds up to 2 gallons!), and use delicate tip muscles to pick up a single berry. It's a snorkel, a hand, a drinking straw, and a hose all in one.

 **Eats:** leaves, fruit, bark, grass — up to 300 lbs per day!

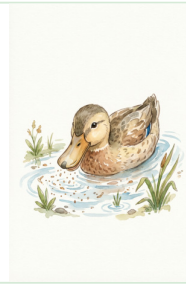


Giant Anteater

Long snout + long, sticky tongue

The narrow **snout** pokes into termite mounds too small for predators to reach. The **tongue** (up to 24 inches long!) is covered in sticky saliva and flicks in and out up to 160 times per minute, collecting thousands of termites per visit.

 Eats: up to 35,000 ants and termites per day



Duck

Flat bill with comb-like edges (lamellae)

The **flat bill** scoops water. Tiny comb-like structures called **lamellae** along the bill's edge act like a strainer — water squirts out but small plants, seeds, and invertebrates stay trapped inside. This is called filter feeding.

 Eats: aquatic plants, small invertebrates, seeds



Giraffe

Long neck + prehensile lips and tongue

The **long neck** reaches leaves 18 feet off the ground — higher than any other browsing animal. The **prehensile lips and tongue** (up to 18 inches long, dark blue to resist sunburn!) wrap around thorny acacia branches and strip the leaves clean.

 Eats: acacia leaves that are toxic to most animals (giraffes tolerate the toxins)



Woodpecker

Chisel beak + long barbed tongue

The **chisel beak** hammers into tree bark to expose insect tunnels — at up to 20 pecks per second! The **barbed, sticky tongue** (so long it wraps around the skull!) reaches into the tunnel to extract insects. The skull has special shock-absorbing structures to prevent brain injury from all that pecking.

 Eats: wood-boring beetles, ants, and tree sap

Discussion during Part 1:

- "Could an eagle eat like a duck? Could a duck eat like an eagle? Why not?"
- "Which animal has the most impressive food-getting structure? Why?"
- "What tool do YOU use to eat? (Fork, spoon, fingers, teeth) — how do those compare to animal tools?"



Part 2 — Draw an Animal Eating (15 min)



Drawing Challenge: Show How It Eats!

Pick *one* animal from above. Draw a scene showing that animal using its specialized body part to get its food. Your drawing must show:

- The animal
- The specific body structure being used (beak, trunk, tongue, etc.)
- The food being obtained
- At least one label explaining how the structure helps



Discussion Question



Big Question: "If a woodpecker's beak broke, what would happen?"

Suggested answer: The woodpecker would likely starve, because its entire feeding strategy depends on that specialized beak. This is the risk of overspecialization — an animal with a very specific tool for a very specific food is in serious trouble if that tool breaks or that food disappears. The giant panda is a famous example: it eats almost nothing but bamboo, and its "thumb" (actually an enlarged wrist bone) evolved specifically for holding bamboo stalks. If bamboo disappears from a region — which can happen when bamboo forests die — pandas starve. More generalist animals like raccoons, crows, and humans can eat almost anything and are much more adaptable when food sources change. Specialization is a tradeoff: you become very good at one thing, but vulnerable if that one thing is threatened.



Extensions

-  **Beak Challenge (Kitchen Science):** Set up "food sources" — rice in water (duck filtering), marshmallows to skewer (eagle gripping), small objects in a hole in a foam cup (anteater extracting). Try each with tweezers (eagle), a spatula (duck), and a straw or toothpick (anteater). Which tool is best for each food?
-  **Giraffe's Secret:** Acacia thorns are up to 3 inches long — yet giraffes eat them! Their leathery lips and long tongue navigate between thorns. Look up a video of a giraffe eating to see this in action.
-  **Woodpecker Skull:** Search "woodpecker skull shock absorbers" — the skull has a spongy bone structure, a thick beak base, and the tongue literally wraps around the skull to cushion impact. A remarkable structural solution!
-  **Read Aloud:** "What Do You Do With a Tail Like This?" by Steve Jenkins (features many food-getting structures!)
-  **Generalist vs. Specialist:** Compare what a raccoon can eat vs. what a koala can eat (only specific eucalyptus leaves). Which would survive better if their environment changed? Discuss the tradeoffs.



Parent/Teacher Notes

- **The woodpecker discussion:** The overspecialization concept is sophisticated but accessible. The core idea for Grade 1: "Being really good at one thing means you depend on that one thing." You can connect it to human specialization too — a surgeon is very specialized; a general doctor sees many things. What are the tradeoffs?
- **Elephant trunk science:** An elephant's trunk has about 40,000 muscles (humans have about 600 total). It can pick up a log weighing 700 lbs or pick a single blade of grass. It can suck up 2 gallons of water and squirt it into the mouth — or use it as a snorkel when swimming. It's one of the most versatile body parts in the animal kingdom.
- **Anteater tongue detail:** Anteaters have no teeth at all. They swallow small stones along with their insect prey — the stones grind up the insects in their stomach. The tongue itself produces so much saliva it runs down the snout in streams.
- **Duck lamellae:** The comb-like lamellae along a duck's bill are analogous to baleen whale plates — both are filter-feeding structures at very different scales. This is a beautiful example of convergent evolution (different animals evolving similar solutions).
- **Kitchen beak challenge tip:** This works wonderfully with: tweezers = eagle/heron (gripping fish), spoon = duck (scooping), straw or long toothpick = anteater (probing), clothespin = parrot (cracking seeds). Use rice, small marshmallows, dried beans, and a bowl of water as your "food sources."
- **Giraffe toxin tolerance:** Acacia leaves contain tannins that are unpleasant or toxic to most herbivores. Giraffes have evolved tolerance to these toxins — another example of co-evolution between animal and food source.



Student Worksheet — Structures for Getting Food

Name: _____ Date: _____

Part 1 — Matching: Animal Structure to Its Food

Draw a line from each animal structure to what it helps the animal eat!



Eagle's talons and hooked beak



Elephant's trunk



Anteater's long sticky tongue



Duck's flat bill with comb edges



Giraffe's long neck and lips



Woodpecker's chisel beak

Drill into tree bark to find insects

Reach inside termite mounds

Catch and tear apart meat

Strip leaves from tall treetops

Reach high branches and suck up water

Filter tiny organisms from water

Part 2 — Draw an Animal Eating!

Pick one animal from above. Draw it using its special body part to get its food. Label the body part and the food!

I am drawing: _____



Draw your animal eating here...

The special body part is:

It helps the animal get:

Part 3 — Think About It!

What would happen if a woodpecker's beak broke? Why?



PARENT ANSWER KEY — Detach before giving worksheet to students

Part 1 — Matching Answers



Structure → Food Function



Eagle talons + beak → **Catch and tear apart meat**



Elephant trunk → **Reach high branches and suck up water**



Anteater tongue → **Reach inside termite mounds**



Duck flat bill → **Filter tiny organisms from water**



Giraffe neck + lips → **Strip leaves from tall treetops**



Woodpecker chisel beak → **Drill into tree bark to find insects**



Part 3 — Think About It!

The woodpecker would likely starve

Its entire feeding method depends on its beak — no beak means no way to reach its food

Accept: "it couldn't get into trees to eat" or "its food would be gone"

Bonus: student connects this to overspecialization risk

Part 2 — Drawing Challenge — What to Look For

- Drawing clearly shows the animal's specialized structure being used
- Food source is depicted or described
- Labels identify the body part and the food
- Student can verbally explain how the structure helps



What Success Looks Like

- Student correctly matches all 6 animal structures to their food functions
- Student understands: each structure is suited to a specific food source
- Student can explain the woodpecker answer: specialized tool = dependent on specific food
- Bonus: student can explain what "overspecialization" means and give an example (panda)



Vocabulary bonus: The word *prehensile* means able to grasp or hold. Giraffes have prehensile lips and tongue. Monkeys have prehensile tails. Elephants have a prehensile trunk. See if your student can think of other prehensile body parts! (Your own hands are prehensile — you can wrap your fingers around objects.)

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Structures for Moving

Life Science Unit 1, Lesson 6 — Unit Finale!



Grade: 1st Grade



Duration: 45 minutes



Subject: Life Science — Animal Structures



Lesson: 6 of 6 ★



Water, air, underground, treetops, desert — every environment has an animal perfectly built to move through it!



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **identify** structural adaptations for movement in different environments and **design** an original animal with structures suited to a specific environment.



The Big Idea

Moving around is essential for survival — animals need to find food, escape predators, find mates, and reach shelter. Different environments demand completely different movement strategies. The structures that make a fish an effortless swimmer would be useless underground. The paws that make a mole a perfect digger would slow it down in water. Every environment shapes its animals to move through it efficiently.



Key vocabulary: streamlined, fins, membrane, adaptation, environment, burrow, prehensile, splayed, tundra, habitat

Materials

What You Need

- Paper and crayons or markers (for the animal design challenge)
- Colored pencils — *optional, for more detailed animal designs*

✓ *Paper and crayons — that's all! The creativity is the activity.*

Part 1 — Five Environments, Five Solutions (15 min)

Each environment presents a different movement challenge. Study how animals solved each one.



Water

Fish, dolphin, duck, penguin, sea turtle

Streamlined body — tapered front to back, reduces drag as water flows past

Fins — push against water to steer and propel forward

Paddles (ducks, penguins) — webbed feet act like oars for surface swimming

Flippers (sea turtles, seals) — like wings, "fly" through water



Air

Eagle, bat, flying squirrel, sugar glider

Feathered wings (birds) — rigid yet flexible, generate lift and thrust

Membrane wings (bats) — skin stretched between long arm and finger bones, incredibly maneuverable

Gliding membranes (flying squirrel) — stretched skin between front and back legs acts as a parachute — glides but can't truly fly



Underground

Mole, gopher, naked mole rat, earthworm

Powerful front legs — built for digging, like shovels with muscles

Shovel-shaped paws — broad, flat hands with extra bone (a "sixth finger") to scoop soil

Tiny or no eyes — useless underground, replaced by sensitive nose and whiskers

Compact body — fits through narrow tunnels



Treetops

Spider monkey, sloth, chameleon, tree frog

Grasping hands and feet — opposable thumbs wrap around branches

Prehensile tail (spider monkey) — acts as a fifth arm, wraps around branches for anchor

Sticky toe pads (tree frog) — microscopic suction cups grip any surface

Long arms and legs — reach gaps between branches



Desert Sand

Camel, sand cat, shovel-nosed snake, sandgrouse

Wide, splayed toes (camel) — distribute weight across soft sand, like built-in snowshoes (sand shoes!)

Padded feet (camel) — thick leathery sole protects from burning hot sand

Lateral undulation (sidewinder snake) — moves sideways to minimize contact with hot sand

Sand-colored coat — camouflage and heat reflection in desert light

Discussion during Part 1:

- "Could a mole survive in the ocean? Why not?"
- "Birds and bats both fly — how are their wings different? Same job, different structure!"
- "What do a spider monkey's tail and a rock climber's rope have in common?"

Hands-On Test — Water Resistance (5 min): Fill a bowl with water. Try pushing a flat piece of cardboard through it, then a pointed cone shape. Ask: "Which moves more easily?" Connect that back to why fish and dolphins have streamlined bodies for moving through water.

Hands-On Test — Prehensile Tail Challenge (3 min): If safe, have students hang from their arms for 30 seconds and notice how hard their muscles work. Then connect that feeling back to spider monkeys using a powerful tail like an extra gripping tool while moving through treetops.

Part 2 — Design Your Own Animal (20 min)

Engineering Challenge: Invent an Animal!

Pick *one* environment for your animal to live in:



Deep Ocean



Dense Jungle



Frozen Tundra



Hot Desert

Draw your made-up animal and label *at least three* structural adaptations that help it move through that environment. For each one, explain *why* that structure helps in that environment.

Bonus challenge: Give your animal a name and explain what it eats and what it must escape from!

Step-by-Step Guide for Part 2:

1. Review the five environments and their movement solutions together.
2. Student picks one environment. Discuss: what challenges does movement in that environment have?
3. Student draws an original animal designed for that environment.
4. Label at least 3 structural adaptations for movement — with a reason for each.
5. Share with the family: "My animal lives in _____. It can move there because _____."

Discussion Question

 **Big Question:** "Birds and bats both fly — but their wings are completely different. How can two animals solve the same problem in different ways?"

Suggested answer: This is called convergent evolution — different animals, starting from different ancestors, arriving at similar solutions to the same problem. Bird wings evolved from scaly dinosaur arms; bat wings evolved from mammal hands with skin stretched between elongated fingers. They solve the same problem (powered flight) with different structural approaches. Bird wings use stiff, lightweight feathers that can be rearranged for different flight modes. Bat wings use flexible skin membranes that make bats extraordinarily maneuverable at low speeds — perfect for catching moths in cluttered spaces. Nature doesn't have just one answer to "how do you fly?" — it has many! This same principle shows up everywhere: eyes evolved independently in vertebrates and in octopuses, even though they look remarkably similar. When the same solution keeps appearing independently, it's a sign that it's a very good solution to a real problem.



Extensions

-  **Bat vs. Bird Wings:** Look up pictures of a bird wing skeleton and a bat wing skeleton side by side. Both are modified forelimbs — both have a "shoulder," "elbow," and "wrist" — but the finger bones in bats are elongated to support the wing membrane. Extraordinary!
-  **Read Aloud:** *"Move!"* by Steve Jenkins and Robin Page (features exactly this topic!)
-  **Tundra Animal Research:** The Arctic fox has fur on the bottom of its paws — the only dog family member with this feature. Why? Hot pavement burns paws; frozen tundra freezes them. The foot-fur is insulation for walking on ice. How else do Arctic animals adapt for movement in frozen environments?



Parent/Teacher Notes

- **Convergent evolution note:** The "birds and bats" discussion opens the door to convergent evolution. This is a rich idea for curious students: the same selective pressure (needing to fly, needing to swim fast) tends to produce similar solutions. Dolphins and sharks look similar but are not closely related — both evolved streamlined bodies for fast swimming. Shrews and moles look similar but are only distantly related — both evolved compact burrowing bodies. The same environment pulls animals toward similar solutions.
- **Mole anatomy:** Moles have a remarkable extra bone in their front paws — technically a sesamoid bone in the wrist that functions like a sixth finger, dramatically increasing the scoop surface. This is a beautiful example of how evolution can produce novel structures to solve specific problems.
- **Flying squirrel vs. bat distinction:** Flying squirrels don't truly fly — they glide. They can't gain altitude; they can only extend a downward trajectory. Bats can truly fly, gaining altitude by flapping. This is a meaningful distinction for curious students.
- **Design challenge freedom:** The "make your own animal" activity should be open-ended and celebrated. There's no wrong answer as long as the structural adaptations are logically connected to the environment. A student who designs a deep-ocean creature with bioluminescent stripes for seeing in the dark and webbed feet for swimming is doing real scientific thinking!
- **Desert sand note:** A camel's foot pad covers an area roughly equivalent to a dinner plate. This distributes the camel's 1,500 lb weight so its pressure per square inch on sand is actually lower than a human's feet. Truly remarkable engineering!



Student Worksheet — Structures for Moving

Name: _____ Date: _____

Part 1 — Matching: Animal to Its Movement Environment

Draw a line from each animal to where it moves best!



Fish (streamlined body + fins)



Eagle (feathered wings)



Mole (shovel paws)



Spider monkey (prehensile tail)



Camel (wide splayed toes)



Treetops



Desert sand



Water



Air



Underground

Part 2 — Design Your Animal!

Pick an environment and design an animal built to move in it. Label at least 3 adaptations!

My environment is: _____ My animal's name: _____



Draw your original animal here...

Adaptation 1:

Adaptation 2:

Adaptation 3:

Why it helps:

Why it helps:

Why it helps:

Part 3 — Think About It!

A bird and a bat both fly. How are their wings different?



PARENT ANSWER KEY — Detach before giving worksheet to students

Part 1 — Matching Answers



Animal → Environment



Fish → **Water**



Eagle → **Air**



Mole → **Underground**



Spider monkey → **Treetops**



Camel → **Desert sand**



Part 3 — Think About It!

Bird wings have *feathers*; bat wings have *skin membranes*

Bat wings are stretched between finger bones; bird wings use stiff flight feathers

Accept: "bats have skin between their fingers, birds have feathers"

Both work for flying — different structures, same function

Part 2 — Design Challenge — What to Look For

- Animal clearly lives in the chosen environment
- At least 3 structural adaptations are labeled
- Each adaptation has a reason connected to movement in that environment
- Student can verbally explain their animal's design choices
- Designs are evaluated on *logic of connection* between structure and function — not artistic quality



Unit Completion — What Success Looks Like

- Student correctly matches all 5 animals to their environments
- Student's design shows 3+ labeled adaptations with reasons
- Student can explain: "birds and bats both fly but have different wing structures"
- Student can make the connection across the unit: *structure* → *function* → *survival*



Unit reflection conversation: Ask your student: "If you could have any one animal superpower from this whole unit — a shell, quills, photosynthesis, echolocation, a prehensile tail — which would you pick and why?" There's no right answer — just great thinking about structure and function!



Little Lab Coats



Structures Have Jobs

PROTECTION • MOVEMENT • GETTING FOOD •
ATTRACTION



PROTECTION

Keep animals safe



Hard shell blocks bites



Spines say "stay away"



Camouflage = hiding in plain sight



GETTING FOOD

Specialized tools



Eagle talons grip fish



Long trunk reaches leaves



Sticky tongue snaps bugs



NO PROTECTION

Easy prey for predators



NO FOOD STRUCTURES

Can't catch food



NO MOVEMENT

Stuck in one spot



NO ATTRACTION

No pollination



Chameleons
change color to
communicate



Some flowers
only open at night
for moths



Sharks have
3,000 teeth in a
lifetime



Spider silk is
stronger than steel
wire

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Plant & Animal Structures Quiz

Life Science · Unit 1 · Grade 1

QUESTION 1 OF 10



Engineers who get their best ideas from nature are using a technique called what?

Photosynthesis

Biomimicry

Camouflage

Pollination

QUESTION 2 OF 10



What do ROOTS do for a plant?

Make food using sunlight

Attract pollinators with color

Anchor the plant in the ground and absorb water

Make seeds for new plants

QUESTION 3 OF 10



Leaves are the food factories of a plant. What do leaves use to make food?

Soil and worms

Sunlight, water, and air (carbon dioxide)

Nectar and pollen

Rain and wind

QUESTION 4 OF 10

 **A turtle has a hard shell on its back. What type of animal DEFENSE does this represent?**

Camouflage — blending in with surroundings

Warning colors — bright colors to say 'I'm dangerous!'

Hard shell armor — a rigid cover to protect soft body parts

Spines and quills — sharp points to poke predators

QUESTION 5 OF 10

 **A flower is bright red with a long narrow tube shape and nectar hidden deep inside. Which pollinator is this flower most likely trying to attract?**

A bee (bees love blue/purple and flat landing spots)

A hummingbird (can see red and has a long beak to reach deep nectar)

A fly (attracted to the smell of rotting meat)

A bat (active at night, attracted to pale flowers)

QUESTION 6 OF 10

 **An eagle has sharp curved TALONS and a hooked BEAK. What are these structures best suited for?**

Filtering tiny plankton from water

Cracking open hard seed shells

Catching and tearing apart prey like fish and rabbits

Drilling holes into tree bark to find insects

QUESTION 7 OF 10

 **A hedgehog is covered in sharp spines. When a fox tries to bite it, the hedgehog curls into a ball. How do the spines PROTECT the hedgehog?**

They help it blend into tree bark to hide

They store food for the winter

They make it painful for predators to bite or grab it

They attract mates with a beautiful display

QUESTION 8 OF 10

 **Bees can see the color BLUE and PURPLE very well. Which flower structure uses this fact to attract bees?**

A deep red tube-shaped flower with nectar far inside

A wide, flat, blue or purple flower with a landing platform

A pale white flower that blooms at night with a strong smell

A flower that smells like rotting meat

QUESTION 9 OF 10

 **A fish has a streamlined (smooth, torpedo-shaped) body and FINS. A bat has MEMBRANE WINGS made of skin stretched between long finger bones. What do these structures have in common?**

Both are used for defense against predators

Both are adaptations that help animals MOVE through their environment

Both are used to attract mates

Both help the animal find food underground

QUESTION 10 OF 10

 A woodpecker has a chisel beak for drilling into trees AND a long, barbed sticky tongue for catching insects inside. A panda has a special 'thumb' (actually a wrist bone) just for gripping bamboo. What do these examples show?

All animals eat the same foods

Animals with specialized structures for getting food are always the strongest

Animals' body structures are matched to the specific food they eat

Structure and function have nothing to do with each other

Plant & Animal Structures Unit Quiz Answer Key

Little Lab Coats • Grade 1 • Parent/Teacher Copy

1. 🦉 Engineers who get their best ideas from nature are using a technique called what?

Answer: B. Biomimicry

Biomimicry means copying nature's ideas to solve human problems! For example, duck webbed feet inspired swim fins, and cat retractable claws inspired folding tools. Nature has been solving problems for billions of years!

2. 🌱 What do **ROOTS** do for a plant?

Answer: C. Anchor the plant in the ground and absorb water

Roots do two important jobs: they anchor the plant in the soil so it doesn't fall over, and they absorb water and nutrients from the ground. Tiny tubes in the roots suck up water like a straw!

3. 🌿 Leaves are the food factories of a plant. What do leaves use to make food?

Answer: B. Sunlight, water, and air (carbon dioxide)

Leaves capture sunlight and use it — along with water from the roots and carbon dioxide from the air — to make sugar (food). This amazing process is called photosynthesis!

4. 🐢 A turtle has a hard shell on its back. What type of animal **DEFENSE** does this represent?

Answer: C. Hard shell armor — a rigid cover to protect soft body parts

A hard shell is armor! When danger comes, the turtle can pull its head and legs inside — it's like carrying a shield AND a house. The shell is made of bone and is very hard to bite through.

5. 🌺 A flower is bright red with a long narrow tube shape and nectar hidden deep inside. Which pollinator is this flower most likely trying to attract?

Answer: B. A hummingbird (can see red and has a long beak to reach deep nectar)

Hummingbirds can see the color red (bees can't!), and they have long beaks perfect for reaching nectar deep in tube-shaped flowers. The flower's color, shape, and nectar location evolved together to attract exactly the right pollinator!

6. 🦅 An eagle has sharp curved **TALONS** and a hooked **BEAK**. What are these structures best suited for?

Answer: C. Catching and tearing apart prey like fish and rabbits

Eagle talons grip prey so tightly even a struggling fish can't escape, and the hooked beak tears meat into pieces small enough to swallow. These structures are perfectly matched to hunting other animals for food.

7. 🦔 A hedgehog is covered in sharp spines. When a fox tries to bite it, the hedgehog curls into a ball. How do the spines PROTECT the hedgehog?

Answer: C. They make it painful for predators to bite or grab it

Spines and quills are a pain-based defense! A predator that tries to bite gets poked — sometimes the quills have tiny backwards hooks that make them even harder to remove. The hedgehog curls up to protect its soft belly.

8. 🐝 Bees can see the color BLUE and PURPLE very well. Which flower structure uses this fact to attract bees?

Answer: B. A wide, flat, blue or purple flower with a landing platform

Blue and purple flowers attract bees because bees can see those colors best! Bees also need flat, wide flowers to land on — like a little helicopter pad. Bees even see UV patterns on flowers that humans can't see at all!

9. 🐟 A fish has a streamlined (smooth, torpedo-shaped) body and FINS. A bat has MEMBRANE WINGS made of skin stretched between long finger bones. What do these structures have in common?

Answer: B. Both are adaptations that help animals MOVE through their environment

Both are movement adaptations! A fish's streamlined body and fins help it glide through water with almost no resistance. A bat's membrane wings make it incredibly maneuverable in the air. Different environments, different structures — same goal: move efficiently!

10. 🦊 A woodpecker has a chisel beak for drilling into trees AND a long, barbed sticky tongue for catching insects inside. A panda has a special 'thumb' (actually a wrist bone) just for gripping bamboo. What do these examples show?

Answer: C. Animals' body structures are matched to the specific food they eat

Each animal's body parts are perfectly matched to what it eats! The woodpecker's beak and tongue are specialized tools for one type of food. The panda's 'thumb' works only for bamboo. Specialization makes them very efficient — but means they're in trouble if that food disappears.